# Coronary Artery Calcification and Animal Fluency Performance in Cognitively Healthy Older Adults

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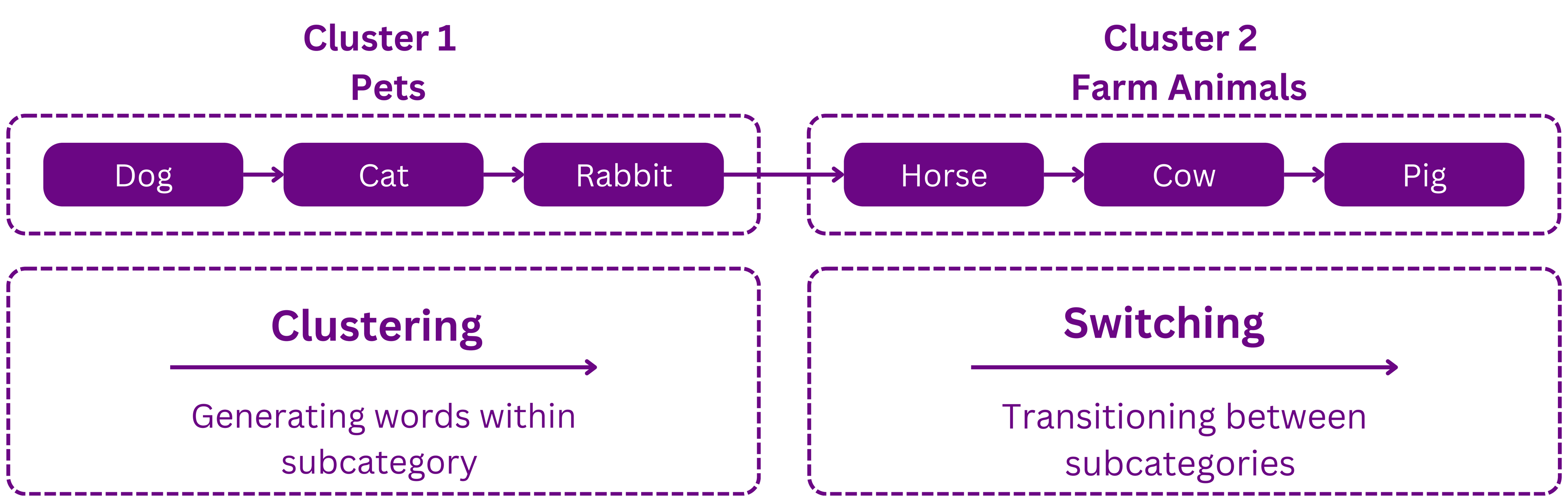
## Introduction

- Coronary artery calcification (CAC) is an imaging biomarker of atherosclerosis, linked to increased dementia risk and poorer cognition (Bos et al., 2012).
- Specific relationship between animal fluency and CAC is unknown.
- Animal fluency tests are sensitive to early neurodegeneration (Wright et al., 2023).
- Reliance on total number of correct words in animal fluency may miss underlying strategies (Troyer et al., 1997).
- Insight into mean cluster size & number of switches may be useful to detect early cognitive change (Zemla et al., 2020).

**Figure 1.** Schematic representation of clustering and switching




## Research Objective

To explore whether cognitively healthy older adults with high CAC (score ≥ 300) differ in total words produced, mean cluster size, and number of switches during an animal fluency task in comparison to older adults with absent CAC (score = 0).

## Methods

- 275 cognitively healthy older adults (MemoLife; Marcolini et al., 2026).
- CAC assessed via low-dose cardiac CT.
- Mean cluster size and number of switches extracted using Dutch animal fluency scheme of SNAFU (Rofes et al., 2023; Zemla et al., 2020).
- Three separate ANCOVAs were run with age, gender, & education, as covariates.

## Results

**Table 1.** Differences in animal fluency metrics between two CAC groups

| Animal fluency metric | CAC = 0<br>n = 149 | CAC ≥ 300<br>n = 126 | Total N = 275 | p | p-FDR |
|---|---|---|---|---|---|
| **Total correct words** | 25.70 (5.88) | 23.80 (5.50) | 24.81 (5.77) | 0.099 | 0.29 |
| **Number of switches** | 9.06 (3.06) | 8.01 (2.58) | 8.58 (2.90) | 0.193 | 0.29 |
| **Mean cluster size** | 2.64 (0.84) | 2.72 (0.79) | 2.68 (0.82) | 0.784 | 0.784 |

***Note.*** Mean (SD) reported for all outcomes. Adjusted means from ANCOVA controlling for age, sex, years of education. FDR = false discovery rate correction (Benjamini–Hochberg method).

## Discussion

- **No significant difference** between older adults with high CAC and absent CAC on the animal fluency metrics.
- Animal fluency abnormality may occur at later MCI and dementia stages (Öhman et al., 2026).
- Insignificant finding for mean cluster size consistent with preserved semantic organization, even in MCI (Wright et al., 2023).
- Significant effects of age and education as covariates highlight the importance of demographic factors.
- White-matter correlates of these metrics were explored as part of the next analysis step.
- Future directions: continuous CAC scores, assessing letter fluency, & adopting item-level approach.

## Code and Scripts

Please scan the QR code to view the scripts for the animal fluency data and the ANCOVA analysis.

## References

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